

D490

Cybersecurity Graduate Capstone

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A graduate student at a regional bank with 60 branches proposes a capstone to “improve cybersecurity at the bank.” Her manager, who holds two security certifications, says the idea is too broad. Which revision best makes it a workable capstone problem?

- A. Pick a topic that matches the manager's certifications.
- B. Narrow it to one evidenced problem, such as reducing phishing-driven account takeovers in the call center, with defined scope and success measures.
- C. Cover every security domain so the project shows full program integration.
- D. Replace the project with a literature review of security trends.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A capstone plan for a 300-bed hospital adds 24/7 log monitoring with alerts on unusual access to patient records. The hospital opened a new wing last year. Under the NIST Cybersecurity Framework, which function does this control mainly support?

- A. Protect.
- B. Respond.
- C. Detect.
- D. Identify.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which activity belongs to the Recover function of the NIST Cybersecurity Framework?

- A. Building an inventory of hardware assets.
- B. Installing endpoint protection on laptops.
- C. Isolating an infected host from the network.
- D. Restoring systems from clean backups and reporting restoration status to stakeholders.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A capstone team at a school district lists three findings: an internet-facing student portal with a critical unpatched flaw that is being actively exploited elsewhere; outdated password posters in the staff lounge; and a lab printer with default settings on an isolated network. The district's superintendent was hired in July. Which should the plan address first?

- A. The staff-lounge posters, because they are the quickest fix.
- B. All three at the same time with equal effort.
- C. The lab printer, because default settings are always the most serious finding.
- D. The student portal.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Two risks each have an impact rating of 4 on a 1–5 scale. Risk A has a likelihood of 2; Risk B has a likelihood of 5. Using $\text{risk} = \text{likelihood} \times \text{impact}$, which statement is correct?

- A. Risk B scores 9 and Risk A scores 6.
- B. Neither can be ranked without a dollar figure.
- C. Risk B scores 20 and outranks Risk A at 8.
- D. They are equal because their impact is the same.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A capstone objective reads: “Make employees better at security.” The organization is a 900-person engineering firm that won an industry award last year. Which rewrite meets the SMART standard?

- A. Cut the phishing-simulation click rate from 18% to under 5% across all staff within six months.
- B. Win another industry award for security culture.
- C. Make every employee fully security-aware.
- D. Run security awareness training for employees.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A CISO at an online retailer with a new logo tracks two numbers: the percentage of critical patches applied within 14 days, and the count of internet-facing systems with known critical flaws. The marketing team recently rebranded the site. Which of these is a key risk indicator (KRI)?

- A. The patch percentage, because it measures the security team's work.
- B. The count of internet-facing systems with known critical flaws.
- C. The number of pages changed in the rebrand.
- D. Neither, because KRIs must always be stated in dollars.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which metric is a key performance indicator (KPI) for a security awareness program rather than a key risk indicator?

- A. Number of accounts compromised through phishing this quarter.
- B. Count of users holding administrator rights.
- C. Any security metric, since every metric is both.
- D. Percentage of staff completing training on schedule.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

An online ordering system for a food distributor with 14 trucks can lose no more than 1 hour of order data and must be back online within 4 hours of an outage. Its servers were replaced in 2023. At minimum, how often must backups or replication capture the data?

- A. Every 4 hours, set by the RTO.
- B. At least every hour, set by the 1-hour RPO.
- C. Once a night, which is standard for all businesses.
- D. Every 5 hours, the RTO plus the RPO.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A payroll system has a recovery time objective (RTO) of 8 hours. What does that figure set?

- A.** The longest the system may be down before service is restored.
- B.** The minimum hours the system must stay up each day.
- C.** That backups must run every 8 hours.
- D.** The most data, measured in time, that may be lost.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A capstone proposes buying a new email-filtering tool for a law firm with offices in two cities. The managing partner likes the vendor's dashboard. Reviewers say the plan will not last. What is most likely missing?

- A.** Nothing; a strong tool by itself sustains a program.
- B.** A more detailed dashboard.
- C.** The people and process parts: an owner, policy, training, and regular reviews that keep the control working.
- D.** A second filtering tool from a different vendor.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A capstone reports that a company has an incident response plan, approved in March and posted on the intranet. The company sponsors a local 5K race. What evidence best shows the plan is effective?

- A.** Results from a tabletop exercise or test showing the team followed the plan and met response targets.
- B.** The plan is effective because approval always proves effectiveness.
- C.** The plan's approval date and signatures.
- D.** The company's community sponsorships.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A graduate student at a regional bank ~~with 60 branches~~ proposes a capstone to “improve cybersecurity at the bank.” Her manager, ~~who holds two security certifications~~, says the idea is too broad. Which revision best makes it a workable capstone problem?

Target word: workable problem statement **Decides:** The proposal lacks a specific, bounded problem.

A	Camouflage	The manager's credentials do not define the organization's problem.
B	Correct	A specific problem with scope and measures is workable and testable.
C	Absolute language	Makes the scope even broader.
D	Adjacent category	A review is not a solution to an organizational problem.

2. Answer C Concept 2

A capstone plan for a 300-bed hospital adds 24/7 log monitoring with alerts on unusual access to patient records. ~~The hospital opened a new wing last year.~~ Under the NIST Cybersecurity Framework, which function does this control mainly support?

Target word: CSF function **Decides:** Monitoring and alerting spot events as they happen.

A	Adjacent category	Protect covers safeguards that prevent events, such as access control.
B	Confusion set	Respond is what happens after an alert, such as containment.
C	Correct	Continuous monitoring and alerting are detection activities.
D	Confusion set	Identify covers asset inventory and risk assessment.

3. Answer D Concept 2 · transfer

Which activity belongs to the Recover function of the NIST Cybersecurity Framework?

Target word: activity from the function — reverse **Decides:** Recover restores capabilities after an incident.

A	Inverted operation	Asset inventory is Identify, the start of the cycle, not recovery at the end.
B	Confusion set	A preventive safeguard under Protect.
C	Adjacent category	Containment is a Respond activity.
D	Correct	Restoration and recovery communication are Recover activities.

4. Answer D Concept 3

A capstone team at a school district lists three findings: an internet-facing student portal with a critical unpatched flaw that is being actively exploited elsewhere; outdated password posters in the staff lounge; and a lab printer with default settings on an isolated network. ~~The district's superintendent was hired in July.~~ Which should the plan address first?

Target word: first priority **Decides:** The portal has the highest likelihood and impact.

A	Inverted operation	Ranks by ease instead of by risk.
B	Dropped element	Drops prioritization entirely.
C	Absolute language	An isolated printer has low exposure.
D	Correct	Internet-facing, critical, and actively exploited: the highest likelihood × impact.

5. Answer C Concept 3 • transfer

Two risks each have an impact rating of 4 on a 1–5 scale. Risk A has a likelihood of 2; Risk B has a likelihood of 5. Using risk = likelihood × impact, which statement is correct?

Target word: rank two risks by score **Decides:** $A = 2 \times 4 = 8$; $B = 5 \times 4 = 20$.

A	Inverted operation	Adds the ratings instead of multiplying.
B	Absolute language	Qualitative scores can rank risks without dollars.
C	Correct	Same impact, higher likelihood, higher score.
D	Dropped element	Drops likelihood from the formula.

6. Answer A Concept 4

A capstone objective reads: “Make employees better at security.” The organization is a 900-person engineering firm ~~that won an industry award last year~~. Which rewrite meets the SMART standard?

Target word: SMART objective **Decides:** Needs a measure, a target, and a deadline.

A	Correct	Specific, measurable, achievable, relevant, and time-bound.
B	Camouflage	Borrows the award detail; not tied to a measurable security outcome.
C	Absolute language	Not measurable or achievable, and no deadline.
D	Dropped element	An activity with no measure or time frame.

7. Answer B Concept 5

A CISO at an online retailer ~~with a new logo~~ tracks two numbers: the percentage of critical patches applied within 14 days, and the count of internet-facing systems with known critical flaws. ~~The marketing team recently rebranded the site.~~ Which of these is a key risk indicator (KRI)?

Target word: pick the KRI **Decides:** Exposed critical systems signal risk exposure.

A	Confusion set	Measuring process performance makes it a KPI.
B	Correct	It directly signals rising or falling exposure.
C	Camouflage	A marketing detail, not a security risk measure.
D	Absolute language	KRIs can be counts, rates, or trends.

8. Answer D Concept 5 • transfer

Which metric is a key performance indicator (KPI) for a security awareness program rather than a key risk indicator?

Target word: pick the KPI — reverse **Decides:** A KPI measures how well the program runs.

A	Inverted operation	Signals exposure and harm, which makes it a KRI.
B	Confusion set	A measure of exposure, not of the training program's performance.
C	Absolute language	The two measure different things.
D	Correct	Measures program performance.

9. Answer B Concept 6

An online ordering system for a food distributor ~~with 14 trucks~~ can lose no more than 1 hour of order data and must be back online within 4 hours of an outage. ~~Its servers were replaced in 2023.~~ At minimum, how often must backups or replication capture the data?

Target word: backup frequency **Decides:** RPO = 1 hour of data loss.

A	Confusion set	RTO limits downtime, not data loss.
B	Correct	Backup frequency must not exceed the allowed data loss.
C	Absolute language	A nightly backup could lose up to 24 hours of orders.
D	Inverted operation	Adds two unrelated limits.

10. Answer A Concept 6 · transfer

A payroll system has a recovery time objective (RTO) of 8 hours. What does that figure set?

Target word: meaning of RTO **Decides:** RTO is about downtime.

A	Correct	RTO is the maximum tolerable downtime for recovery.
B	Inverted operation	Turns a downtime limit into an uptime target.
C	Adjacent category	Backup frequency comes from the RPO.
D	Confusion set	That is the RPO.

11. Answer C Concept 7

A capstone proposes buying a new email-filtering tool for a law firm ~~with offices in two cities.~~ ~~The managing partner likes the vendor's dashboard.~~ Reviewers say the plan will not last. What is most likely missing?

Target word: what makes a program durable **Decides:** The plan is technology only.

A	Absolute language	Tools decay without ownership and review.
B	Camouflage	Answers the partner's preference, not durability.
C	Correct	Durability comes from people and process around the tool.
D	Adjacent category	More technology still leaves no owner or process.

12. Answer A Concept 8

A capstone reports that a company has an incident response plan, ~~approved in March and posted on the intranet.~~ ~~The company sponsors a local 5K race.~~ What evidence best shows the plan is effective?

Target word: evidence of effectiveness **Decides:** Effectiveness requires a test of the plan in action.

A	Correct	Testing shows the control works, not just that it exists.
B	Absolute language	Approval is not a performance test.
C	Confusion set	Proves the plan exists, not that it works.
D	Camouflage	Unrelated to incident response.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	9	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Scoping the problem	A capstone solves one specific, evidenced problem with defined scope and success measures.	Choosing a broad goal such as “improve security” to show everything learned.	A vague proposal that a reviewer calls “too broad.”
2	Mapping to the NIST CSF functions	Govern, Identify, Protect, Detect, Respond, Recover: each control supports the function that matches its purpose.	Calling any security tool “Protect,” or mixing Respond and Recover.	“Which function does this control support?”
3	Risk-based prioritization	Address the highest risk first, where risk = likelihood × impact, not the easiest or cheapest fix.	Fixing quick wins first, or treating all findings as equal.	A list of findings with different likelihoods and impacts.
4	SMART objectives	Objectives must be specific, measurable, achievable, relevant, and time-bound.	Writing aspirations like “make staff more secure.”	An objective with no number or deadline.
5	KPIs vs. KRIs	A KPI measures how well a process performs; a KRI signals rising risk exposure.	Treating every metric as a KPI.	“Percentage completed on time” vs. “count of exposed systems.”
6	RTO vs. RPO	RPO is the most data loss allowed (it sets backup frequency); RTO is the most downtime allowed (it sets restore speed).	Setting backup frequency from the RTO.	“Can lose no more than X of data” vs. “back online within Y.”
7	Durable program: people, process, technology	A lasting program pairs tools with ownership, policy, training, and review cycles.	Buying a tool and calling the program done.	A plan made of purchases only.
8	Control effectiveness vs. existence	A control is proven by testing it (exercises, assessments, metrics), not by showing it exists.	Accepting an approved document as proof of effectiveness.	“What evidence shows the control works?”

Course scope (public description): The Master of Science in Cybersecurity and Information Assurance (MSCSIA) Capstone project allows learners to demonstrate their capability to establish a durable cybersecurity and information assurance program. The capstone project challenges learners to integrate skills and knowledge from all program domains into one project that addresses a significant real-world cybersecurity problem.